

Davide Marchesan

Gameplay Programmer | Unreal Engine Developer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)



Software engineer with 8+ years of experience in creating and maintaining applications and scalable systems, now focused on gameplay development with Unreal Engine (C++).

TECH SKILLS

Game Development

Unreal Engine & C++, Slate UI, Unity & C#, Gameplay systems, Network & Replication, Blender

Full-stack & Web Development

TypeScript & JavaScript (Node.js, Express), Angular, Flutter (Dart), React (Next.js, Redux), HTML, CSS (preprocessor like SASS), PHP, C#, Python, SQL and NoSQL databases, GraphQL, REST API, WebSockets, Postman, Figma

Cloud & DevOps

AWS (CDK, Lambda, DynamoDB, RDS, SQS, SNS, SES, IAM, EC2, Cognito etc.), CI/CD (AWS CodeBuild and CodePipeline), OAuth 2.0 & JWT, Git, Linux

GAME PROJECTS

Multiplayer Racing Prototype - Unreal Engine 5 (C++)

[DEMO VIDEO](#) | [GITHUB](#)

TrackMania-inspired multiplayer racing game featuring custom physics and in-game track editor.

- Physics: engineered a custom vehicle movement component using a fixed-step simulation to ensure deterministic physics. Implemented specific drifting mechanics based on lateral friction curves and slip-angle calculations, replicating the iconic high-speed handling of TrackMania
- Network and replication: developed a multiplayer framework using Unreal's Network Replication system, managing synchronized match flows and player states
- System engineering: implemented a timing system where lap times are calculated based on simulation steps rather than system clock, ensuring competitive fairness in network play
- In-game tooling: architected a modular in-game track editor with a grid-snapping system. Developed a seamless workflow allowing users to build, instantly test-drive and save custom tracks for immediate use in multiplayer sessions
- Platform integration: integrated the Steam Online Subsystem to handle session creation, matchmaking, and lobby management, facilitating peer-to-peer connectivity

Escape Room Prototype - Unreal Engine 5 (C++)

[DEMO VIDEO](#) | [GITHUB](#)

First-person escape room featuring interactable objects, environmental puzzles and modular gameplay systems.

- Designer-centric tooling: architected and developed a suite of reusable gameplay components, enabling level designers to build intricate puzzles via Data Assets and Blueprints, define and sequence game objectives, victory conditions and tutorial flows without further C++ intervention
- Modular trigger framework: developed a flexible trigger system that enables designers to link environmental events (e.g. reaching a location, interacting with object) to a specific game logic, such as activating checkpoints, spawning tutorials or updating the quest log
- Character abilities: developed a custom Ability system (e.g. specialized movement or vision mode), ensuring clean separation of concerns between input handling and gameplay effects
- AI: designed and implemented a custom AI system (line of sight, hearing) for basic enemy units, managing state transition between patrolling, chasing and returning to origin

(Work in progress) Flight Prototype with procedural elements - Unreal Engine 5 (C++)

[GITHUB](#)

Flight simulator with procedural map and obstacles inspired by Pandora's floating mountains from Avatar movie.

- Procedural mesh generation: architected a custom tool to generate organic floating structures using Signed Distance Fields (SDF) and Perlin noise functions. Implemented the Marching Cubes algorithm for efficient real-time mesh extraction and polygon generation
- Deterministic world logic (in progress): developing a seed-based generation system to ensure world consistency. Implementing Poisson Disk Sampling for the natural distribution of procedural obstacles, preventing clusters and ensuring optimal spatial spacing across the map
- Flight mechanics (in progress): designing a physics-based flight model using quaternions for smooth, gimbal-lock-free 3D rotations. Implementing custom drag and lift forces to balance arcade responsiveness with realistic flight dynamics
- Performance (in progress): focusing on a seed-driven pipeline to recreate complex environments with minimal memory footprint, optimizing CPU overhead during real-time mesh construction and object placement

PROFESSIONAL EXPERIENCE

NAVOO — Full-stack developer

JANUARY 2017 - DECEMBER 2025

- Architected and deployed scalable serverless ecosystems using AWS, ensuring high availability for multi-tenant platforms while optimizing infrastructure costs
- Led the full-stack development of complex web applications and enterprise dashboards, translating design mockups into high-performance, responsive interfaces using Angular, React and Flutter
- Developed voice-first interactive systems and gaming experiences for the Amazon Alexa Skills Store, managing the entire lifecycle from backend logic to public release
- Streamlined development workflows by implementing robust CI/CD pipelines (AWS CodePipeline), significantly reducing deployment times
- Developed custom internal tools and automation scripts to streamline repetitive development tasks, significantly increasing team productivity and reducing manual deployment overhead
- Collaborated directly with stakeholders to translate business requirements into technical specifications, driving the successful delivery of custom software solutions for diverse clients

Alexa Smart Properties Dashboard - NAVOO

Architected and developed a multi-tenant cloud platform for the hospitality sector, integrated with the Alexa Smart Properties environment.

- Cloud Architecture: engineered a robust, serverless infrastructure using AWS CDK (Infrastructure as Code), leveraging Lambda, DynamoDB, Cognito and API Gateway
- Frontend: designed the frontend architecture using Angular, developing a library of modular and reusable components for a high-performance dashboard that manages real-time voice/visual content and guest analytics
- System integration: integrated Amazon Device Fleet Management and third-party IoT APIs (Shelly) for unified smart-room control, alongside multiple Property Management Systems (PMS) to automate hospitality workflows
- Security and access control: implemented secure Role-Based Access Control and managed identity provider integration via OAuth 2.0 Account Linking and JWT tokens; handled end-to-end subscription logic by integrating Stripe APIs for automated billing
- DevOps: built custom RESTful APIs and established automated CI/CD pipelines

Question of the day - Voice-Based Game (Alexa) - NAVOO

[AMAZON ALEXA SKILLS STORE](#)

- Designed and engineered an interactive voice experience that delivers daily trivia, educational content and real-time audience benchmarks to thousands of users
- Developed the core logic and AWS Lambda integration, ensuring high performance and low-latency responses for a seamless voice-first user journey
- Implemented a scoring feedback system that dynamically compares individual user data against global audience trends
- Managed the full certification and publishing process on the Amazon Alexa Skills Store, maintaining the application through multiple version updates

Question of the day - Flutter game port - NAVOO

- Developed a mobile version of the Alexa game using Dart and Flutter, ensuring feature parity and a consistent UX across iOS and Android
- Successfully integrated the mobile frontend with the existing AWS serverless backend, enabling real-time data synchronization for user scoring and global rankings
- Led the technical feasibility study for multi-platform expansion, demonstrating the viability of unified cloud logic for both voice and mobile interfaces
- Implemented efficient state management and modular UI architecture to facilitate rapid prototyping and future scalability

Waste Collection Assistant - Alexa Skill - NAVOO

[AMAZON ALEXA SKILLS STORE](#)

- Developed a voice-first utility and dedicated web-based control panel to automate and manage complex municipal waste collection schedules
- Built a custom dashboard allowing users to create, edit and synchronize personalized collection calendars, bridging the gap between a web interface and voice responses
- Engineered the backend using AWS Lambda and DynamoDB, ensuring high reliability for time-sensitive notifications and user queries
- Designed the voice interaction model to handle natural language requests, providing accurate, localized information on disposal days and waste categories
- Oversaw the entire development cycle from initial prototype to successful publication and maintenance on the Alexa Skills Store

ADDITIONAL PROJECTS / PROTOTYPES

Waste Collection Assistant - Flutter mobile app - NAVOO

Multi-Platform expansion: designed and developed a cross-platform mobile application using Flutter, seamlessly integrating with the existing AWS serverless backend to provide a unified user experience across voice and mobile touchpoints.

Buzz-style Multiplayer Party Game - Flutter Prototype

[GITHUB](#)

Designed a multiplayer tribute to the PlayStation Buzz series, using AWS API Gateway (WebSockets) to manage bi-directional communication between a TV host interface and multiple smartphone controllers.

Developed a fully serverless backend using AWS Lambda and DynamoDB to orchestrate real-time game states, managing unique Session IDs, player synchronization.

Online Auction Platform – Custom WordPress Plugin - NAVOO

Engineered a custom auction plugin integrated into WordPress, leveraging AWS SQS to manage high volume of concurrent bids, ensuring message durability and reliable transactional processing during peak traffic.

EDUCATION

Game Programmer (1 year)

EVENT-HORIZON, PADUA, ITALY - 2019/2020

Learned the fundamentals of game development working in a team of developers, artists and designers to cooperate and create a game in Unity.

AWS Certified Alexa Skill Builder

AWS - 2019

Web Developer Certificate

DIEFFE, PADUA, ITALY - 2016

High School Diploma - IT & Accounting

CASTELFRANCO VENETO, ITALY - 2015

LANGUAGES

Italian (native)

English (professional proficiency)